

Lesson 6: Is there also a quantitative relationship between the change in momentum, force, and the amount of time the force acts on an object?

Use Mathematical Thinking

Home learning



1. The key provided outlines of multiple approaches for coming to the same conclusion for each question.
2. How can reflecting on our confidence level for using these different approaches help us as learners?

Navigate

With your class



3. What type of tests would we need to carry out to figure out if there is any quantitative relationship between the change in momentum, force, and time?

Considering Sources of Error

Differences in measurements of the same thing indicate errors. Scientists cannot eliminate error entirely from their investigations, but they can take different steps to reduce it.

Turn and talk



4. What are some possible sources of error in our measurements?
5. What could we do to reduce the error in our measurements?
6. If every measurement we collect will have some error:
 - which value(s) should we use?
 - which value(s) should we discard?

Discuss the force(s) acting on the cart

With your class



7. What force(s) do you think are acting on the cart during the collision?
8. How could we modify the force of friction between the bin and the surface?

Discuss the change in momentum of the cart

With your class



9. Discuss the measurements you can take of the system to ensure that the velocity of the cart is the same every time it leaves the ramp?

Considering the Manipulation of Variables

We can change the force acting on the cart **directly** by changing the mass we put on the bin.

We can also **directly** change the momentum of the cart before the collision by changing the mass of the cart.

With a partner



10. Could we make the time of collision longer or shorter without the need of manipulating the other two variables? Why or why not?

- Be ready to share your ideas with the class

Making Predictions

On your own



11. Make predictions for each of the experiments we are going to carry out. Reference the slide or your handout to record predictions.

With a partner



12. Discuss and record some quantitative predictions

- Be ready to share your ideas with the class

Carry out an Investigation

With your group



13. Review the Calibration and Variable Manipulation procedures with your group

14. Carry out the related calibration protocols for your experiment.

15. Carry out your experiment (A, B, or C) and record the results.

16. Use CODAP to analyze your data (Part 4)

Analyzing and Interpreting Data

We wanted to investigate the relationship between the force acting on the cart (average magnitude of stopping force), the time of collision (Δt), and the momentum of the cart.

On your own



17. Pair up with a partner from another group that carried out the same experiment as your group.

18. Work together to complete **Part 5** of the handout to summarize what you both figured out to help answer this question.

- Be ready to share your ideas with the class

Share Main Findings from Investigation

With your class



19. Based on our results from:

- Experiment A: How is the time to stop a moving object with a constant force affected when the momentum of the object doubles?
- Experiment B: How is the time to stop a moving object with a constant mass affected when the magnitude of the force acting on an object doubles?
- Experiment C: How is the time to stop a moving object affected when the momentum of the object doubles AND the magnitude of the average force acting on it doubles?

Represent the Change in Momentum

With your class



20. Develop an area based representation of the data from investigation for one of the conditions you tested.

Use Mathematical Thinking

With a partner



21. Rearrange the 4 variable cards and as many of the gray operation cards as needed into an equation to represent the mathematical relationship that seems to connect these variables.

22. Once you agree with your partner on the mathematical relationship, use the average values obtained from our investigation to test if the two sides of the equation you identified are equal.

- Be ready to share your model with the class

Develop a Mathematical Model

With your class



23. (How) can we describe in words what our proposed mathematical model mean?

24. What would our mathematical model predict for scenarios when an object has a very large momentum at the moment of the collision?

25. How would results from additional collisions help us determine the generalizability and/or limits of this model?

Use a Mathematical Model

On your own



26. Use the mathematical model to make predictions about the time of the collision (Δt) for the following conditions:

- Condition D: How long would it take to stop a 2 kg mass traveling at 1 m/s when a 3 Newton friction force is applied to the cart?
- Condition E: How long would it take to stop a 3 kg mass traveling at 1 m/s when a 4 Newton friction force is applied to the cart?

Analyze additional conditions

With a partner



27. Watch the one video for the condition you were assigned (D or E)

28. Extract the data you need to test whether our proposed mathematical relationship holds for your condition.

- Be ready to share your findings with the class

Discuss the Generalizability of our Model

With your class



29. Did our proposed mathematical model hold for none, one, or both conditions?

30. What does this tell us about the generalizability and/or limits of this model?

Navigate

On your own



31. How can we use our new mathematical relationship to explain why distracted driving might affect the outcomes of a vehicle collision?